

Searching 2D Lists

Look at this code:

```
clubs = [["art", 12], ["chess", 18], ["code", 15]]
target = input("Club to find: ")
target = target.lower()
found = False
for club in clubs:
    if club[0] == target:
        print("Members:", club[1])
        found = True
if found == False:
    print("Club not found.")
```

Use the code above to help you. After completing each challenge below, run your program to test that it works correctly. Save each completed challenge program in a new file.

Programming Challenges

Challenge 1

Change the search so the user enters a member count instead of a club name. Display the matching club. Test 15; the program should display "Club: code".

Challenge 2

When `[club][0]` matches `[target]`, ask for a new member count and store it in `[new_members]`. Update `[club][1]`, then display "Updated:" followed by the changed row. Test "Chess" and 20; display "Updated: ['chess', 20]". Test "Music"; display "Club not found."

Challenge 3

Store `["orbit", 120]`, `["harbour", 96]` and `["forest", 144]` in `[books]`. Ask for a title and store its lower-case value in `[target]`. Use a `for` loop to compare `[target]` with each title. Display "Pages:" followed by the page count when a title matches. Display "Book not found." only after every row has been checked and no match was found. Test "Harbour"; display "Pages: 96". Test "Music"; display "Book not found."

